

Comparison of Vertical Jump Height Using the Force Platform and the Vertec

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Abstract— Vertical jump height is often being measured in many ways to evaluate the ability of an individual to jump, and to provide an estimation of lower limb muscle power. The various jump tests and apparatus each have their advantages and disadvantages. The main purpose of this study was to compare and explore the relationship between the two more popular test apparatus - the force platform and the Vertec. Sixty-nine university students, 32 females and 37 males between the ages of 20 to 23 (M age 22.01 ± 0.66 years) were recruited to participate in this study. They were required to perform countermovement jumps on the force platform and using the Vertec. Both jump tests were run on the same day, with a 10-minute interval between tests. Results showed that the jump height values obtained from the force platform were significantly lower compared to jumping with the Vertec. The mean jump height for the force platform was $0.36\text{m} \pm 0.094$ and for the Vertec was $0.55\text{m} \pm 0.120$ respectively. Although the mean jump height differs significantly, both the force platform and Vertec displayed a significant correlation ($r = 0.826, p < 0.01$). It was concluded that both devices are suitable to be used to measure jump height because those who obtained a high jump height at the Vertec also obtained a high value of jump height on the force platform. An important difference between both devices was the presence of a target of achievement for the Vertec.

Keywords— Jump height, Vertec, Force Platform, Countermovement Jump

1. INTRODUCTION

Generally, every individual has the ability to jump as it is one of the fundamental motor skills that is acquired at a very young age. However, there are many factors that influence the capability of an individual to jump and attain the maximum jump height. The vertical jump test is frequently used to assess athletic skill by comparing the jump performance of starters vs. nonstarters [1] and to measure lower body power [2]. It also appeared to be a valuable measurement in identifying lower-limb functionality for the non-athletic population [3, 4]. While there are many types of devices that can be used to measure vertical jump height, two types of devices, the force platform and the Vertec, was investigated in this study.

A force platform is a rectangular metal plate with a piezoelectric or strain gauge transducer attached at each corner to give an electric output that is proportional to the force on the plate. It is a device that measures the force

applied by an individual onto the force plate. There are few methods that can be used to measure the jump height using the force platform, including the flight time method, impulse-momentum method and the work-energy method [5]. Linthorne [5] found that both flight time and impulse momentum methods obtained via the force platform are able to provide a good estimation of jump height.

Alternately, the Vertec consists of several plastic swivel vanes arranged in half inch (0.0127m) increments which are attached to a telescopic metal pole that can be adjusted to a participants' standing reach. The test requires a participant to use their hand to displace the vanes with an overhead swinging motion at the peak of their vertical jump. The highest displaced horizontal swivel vane determines the maximum jump height. Vertical jump height is calculated by taking the difference between the fully extended standing reach measurement and the highest displaced horizontal swivel vane [6]. The Vertec is relatively inexpensive, require minimal training for operation, do not require extensive data analysis, and provide immediate results. [3]

In previous studies, jump height has been measured using Just Jump, contact mat, and Myotest accelerometers [2, 3, 7, 8, 9, 10, 11]. One study by Erich and colleagues [11] found 27% higher jump height when assessed by the Vertec, compared to the force platform. Another study conducted by Isaacs [7] compared the Vertec to the Just Jump system (VJM) in children aged 7 to 11 years old and found a significant difference ($r = 0.83, p < 0.001$) between vertical jump height scores measured with the VJM and the Vertec. Jump height in the VJM is determined by measuring flight time [3], and the flight time is subsequently determined by microswitches located within the mat, which are sensitive to the lift-off of the feet from the mat and to the landing of the feet back on to the mat [6]. Participants jumping on the mat focuses purely on the jump and do not have the knowledge of results about the height of jump. Meanwhile, jumping using the Vertec would involve the coordination of jumping and swiping the vanes simultaneously as participants have a target to aim and reach. However, previous research has yet to investigate the relationship between the Vertec and the force platform on jump performance. Thus, the purpose of this study is to compare jump height and determine the relationship between the force platform and the Vertec.

II. METHODS

A. Participants

Sixty nine university students, 32 females and 37 males between the ages of 20 to 23 (M age 22.01 ± 0.66 years), from the University of Malaya were recruited to participate in this study. All the participants were healthy and had no injuries during the data collection period. They were briefed on the procedures of the study and required to sign an informed consent form. The study was approved by the University of Malaya Research Ethics Committee (UM.TNC2/RC/H&E/UMREC – 116) and procedures of this study were carried out in accordance with the guidelines regarding the use of human participants.

B. Instruments

The force platform was a rectangle metal piece (0.5m x 0.5m). It was set to 1200Hz for the capture rate. Gen 5 AMTI-NetForce (Watertown, MA, USA) was used in this study. According to Linthorne [5], jump height can be calculated using the flight time method where;

$$v_{to} = \frac{gt_{flight}}{2}$$

$$y_{flight} = \frac{v_{to}^2}{2g}$$

Where V_{to} is the vertical take-off velocity, t flight is duration of participant in air and y_{flight} is the jump height of the participant.

The Vertec (Sports Imports, Columbus, OH, USA) consists of 48 plastic vanes with three colours (white, blue and red). Every red vane indicates 6 inch, blue indicates 1 inch and white indicates 0.5 inch. The difference between standing height and the highest displaced vane measurement is the value of the jump height.

C. Procedures

Participants were required to perform countermovement jumps on two jumping test apparatus, which were carried out in a laboratory on the same day. The rest period between tests was 10 minutes. Height and weight of the participants were recorded. They were briefed on the procedures and the protocols of the tests. Prior to testing, participants warmed up by cycling on an ergometer at an average of 60 RPM for five minutes. Participants then proceeded to either jumping on the force platform followed by jumping with the Vertec, or vice-versa, the order by which was determined randomly. Three trials were given for each jumping device and the

highest score for each was used for analysis. Only countermovement jumps were recorded, and trials with the wrong technique were repeated. A countermovement jump, starts from a standing position, followed by a squat with arms swinging backwards and then taking off the ground with the arms swinging up forward and upwards (see Figure 1 for an illustration).

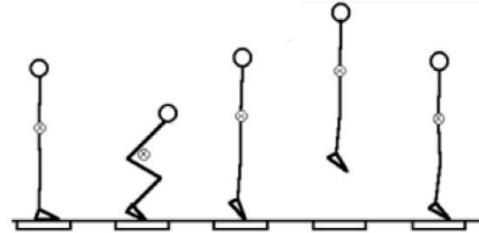


Figure 1. Countermovement Jump
(Adapted from Linthorne [5])

For the force platform jump test, participants stepped on the force platform when instructed and executed a countermovement jump. If they landed outside the platform on descend, the trial was repeated. Rest interval between trials was one minute.

For the Vertec jump test, the plastic vanes were adjusted according to the participants' maximum standing reach by extending their arm straight over the head (standing height). The dominant side was closest to the vanes. Participants then performed a maximal countermovement jump and displaced the vanes with their dominant hand. The highest displaced vane was recorded.

D. Data Analyses

To compare the vertical jump height between jumping on the force platform and jumping with the Vertec, the highest score for each of the jumps on each jumping device was recorded. Data was analyzed using IBM Statistical Package for the Social Science (SPSS) software. Data distribution was assessed using Kolmogorov Smirnov's test for normality. Paired sample t-test was used to determine if there were significant differences between the two types of jump test apparatus with p level set at .01. Relationship between the two tests were analyzed using the Pearson product moment correlation coefficient. Means and standard deviations of jump height (m) were obtained using descriptive statistics.